

DREAM6-ZR Local Boundary Program

Audited checkpoint for an addendum: exact critical-channel entry, numerical regression,
and remaining gates

3 September 2026

Abstract

This note is a status ledger, not a claim of full boundary closure. It separates three logically different layers.

PROVED: within the stated coarse local transport model, the primitive defect is $(1-t)^4/64$, the local $r = 0$ repair debt is $1/32$, the natural local boundary readout has a six-dimensional marginal-kernel quotient, the critical CORE class is positively closed up to a terminal sort, and every hard-edge product boundary state enters CORE through an explicit finite critical-channel transient. The transient is governed by

$$\tau(c) = \max \left\{ 0, \frac{48c - 9}{38} \right\},$$

with landing threshold $13/48$ and exact chain

$$\frac{1}{2} \xrightarrow{x=2} \frac{15}{38} \xrightarrow{x=2} \frac{189}{722} < \frac{13}{48} \xrightarrow{\text{landing}} \text{CORE}.$$

LP-CHECKED: independent floating-point transportation LPs reproduce the constrained constructions on extreme and random finite-support laws. These runs are regression tests only; a PASS is not a Farkas certificate and is not used as a proof step.

OPEN: the $x = 1, 3$ recursive obligations, the depth/fuel semantics of the terminal state t , and the ordered-history lift remain unproved. Consequently neither the full causal addition lemma nor the desired $O(\log n)$ upper bound is claimed here.

1 Extractor-safe statement of the constants

The exact constants used in every subsequent section are:

$$\text{tau}(c) = \max(0, (48*c - 9)/38)$$

$$1/2 \rightarrow 15/38 \rightarrow 189/722 < 13/48 \rightarrow \text{CORE}$$

$$\text{delta}(c) = 23/128 - c/4$$

These plain-text guards are included deliberately so that PDF text extraction cannot silently lose a numerator or denominator.

2 Status ledger

Status	Statement
PROVED	$P_b(t) - Q_b(t) = (1 - t)^4/64$.
PROVED	The two local illegal cells at $r = 0$ carry masses $3/128$ and $1/128$, hence total debt $1/32$.
PROVED	For the stated nine-dimensional local boundary readout, the marginal-kernel image has dimension 6 for every finite support $m \geq 2$.
PROVED	The critical CORE class $\mathcal{C} = \{b_0 = 0, b_1 \leq 13/16\}$ closes on $x \in \{0, 2, 4\}$ up to the distinguished terminal law t .
PROVED	For hard-edge product boundary states, the scalar critical transient satisfies the peeling map $\tau(c)$ above and enters CORE after at most three $x = 2$ steps.
LP-CHECKED	DREAM6-ZR-PHASE v0.2 and the independent entry certifier reproduce the local constructions numerically with explicit residual audits.
OPEN	Incorporation of branches $x = 1, 3$ into a well-founded five-branch induction.
OPEN	A correct remaining-depth/fuel rule for the terminal law t .
OPEN	Lift from the coarse transport to the ordered-history causal phase.
OPEN	Full causal addition lemma and the final $O(\log n)$ consequence.

3 Local transport model

PROVED

The coarse laws are

$$P_b = \frac{1}{64}(1, 12, 38, 12, 1), \quad Q_b = \frac{1}{64}(0, 16, 32, 16, 0),$$

so that

$$\boxed{P_b(t) - Q_b(t) = \frac{(1 - t)^4}{64}.} \quad (1)$$

Let

$$B(t) = \sum_{j \geq 0} b_j t^j, \quad b_j \geq 0, \quad \sum_j b_j = 1,$$

and put

$$q_u = (Q_b * B)_u.$$

A local causal transport is a nonnegative array $\pi(x, u)$ with

$$\sum_u \pi(x, u) = P_b(x), \quad \sum_x \pi(x, u) = q_u, \quad \pi(x, u) = 0 \quad (u < x).$$

The residual law on row x is

$$B_x(j) = \frac{\pi(x, x + j)}{P_b(x)}.$$

At $r = 0$, the natural interior rule creates the two illegal transports

$$(2, 1), \quad (4, 3),$$

with terminal-weighted masses

$$\frac{3}{128}, \quad \frac{1}{128}.$$

Thus the exact local repair bill is

$$\boxed{\frac{3}{128} + \frac{1}{128} = \frac{1}{32}}. \quad (2)$$

The parent-determined donor capacities used below are

$$D_{21}(B) = \frac{13}{64} - \frac{1}{4}b_0, \quad (3)$$

$$D_{43}(B) = \frac{63}{64} - \left(b_0 + \frac{3}{4}b_1 + \frac{1}{4}b_2\right). \quad (4)$$

4 Local phase quotient

PROVED

For the natural nine-coordinate boundary readout

$$(H_1(B_0), H_3(B_0), H_1(B_2), H_3(B_2), H_1(B_4), H_3(B_4), D_{21}, D_{43}, D_{\text{sh}}),$$

three coordinates are already fixed by marginals:

$$H_1(B_0) = 0,$$

$$D_{21} = \frac{13}{64} - \frac{1}{4}b_0,$$

$$D_{43} = \frac{63}{64} - \left(b_0 + \frac{3}{4}b_1 + \frac{1}{4}b_2\right).$$

The remaining marginal-kernel image has exact dimension six. One convenient basis is

$$\Phi = (H_3(B_0), H_1(B_2), H_3(B_2), H_1(B_4), H_3(B_4), D_{\text{sh}}).$$

This is a local statement only: it identifies what the boundary can see inside a transport fiber. It does not assert that a positive phase composes recursively.

5 CORE and terminal sorts

PROVED

Define

$$\mathcal{C} = \left\{ B : b_0 = 0, \quad b_1 \leq \frac{13}{16} \right\}. \quad (5)$$

The terminal sort is the distinguished law

$$\boxed{\mathcal{T} = \{t\}}.$$

The constant 13/16 is an exact Hall-capacity threshold. If $b_0 = 0$, then

$$q_2 = \frac{1}{4}b_1.$$

To force the $x = 2$ child to have zero constant coefficient one needs $\pi(2, 2) = 0$, so column 2 must fit into rows 0, 1 of total capacity 13/64. Hence

$$\frac{1}{4}b_1 \leq \frac{13}{64} \iff b_1 \leq \frac{13}{16}.$$

Proposition 1 (Critical-channel CORE closure – PROVED). *For every $B \in \mathcal{C}$, there exists a causal transport such that*

$$B_0 \in \mathcal{C}, \quad B_2 \in \mathcal{C}, \quad B_4 \in \mathcal{C} \cup \mathcal{T}.$$

The local repair screen attains common margin

$$\delta_{\text{core}} = \frac{23}{128}.$$

Proof. Since $b_0 = 0$,

$$q_0 = q_1 = 0, \quad q_2 = \frac{1}{4}b_1 \leq \frac{13}{64}.$$

Assign column 2 entirely to rows 0, 1, so $\pi(2, 2) = 0$. Hence $(B_2)_0 = 0$. Also

$$q_3 = \frac{1}{2}b_1 + \frac{1}{4}b_2 \leq \frac{1}{4} + \frac{1}{4}b_1 \leq \frac{29}{64},$$

and therefore

$$(B_2)_1 = \frac{\pi(2, 3)}{38/64} \leq \frac{29}{38} < \frac{13}{16}.$$

Thus $B_2 \in \mathcal{C}$. Because columns 0, 1 have zero mass, row 0 has no residual coefficients at 0, 1, so $B_0 \in \mathcal{C}$.

For row 4,

$$F_q(4) = b_1 + \frac{3}{4}b_2 + \frac{1}{4}b_3 \leq \frac{3}{4} + \frac{1}{4}b_1 \leq \frac{61}{64}.$$

Hence at least $3/64$ of column mass lies in $u \geq 5$. If $q_5 \geq 1/64$, place row 4 entirely at $u = 5$, producing $B_4 = t$. Otherwise more than $2/64$ lies in $u \geq 6$, enough to place row 4 entirely there, giving $(B_4)_0 = (B_4)_1 = 0$ and hence $B_4 \in \mathcal{C}$. The remaining columns are at least 4, so all remaining rows 0, 1, 2, 3 can absorb them.

Finally, $D_{21} = 13/64$, so after the $3/128$ debt the first repair inequality leaves $23/128$. The other two repair inequalities have larger slack on $\mathcal{C}$, hence the common maximum is $23/128$. $\square$

6 Scalar entry classes

PROVED

For $0 \leq c \leq 1/2$, define

$$\mathcal{E}(c) = \{B : b_1 = 0, 0 \leq b_0 \leq c\}.$$

The exact peeling map is

$$\tau(c) = \max \left\{ 0, \frac{48c - 9}{38} \right\}. \quad (6)$$

Theorem 1 (Critical peeling step – PROVED). *Let $B \in \mathcal{E}(1/2)$ and let $c = b_0$. There exists a causal transport such that*

$$B_0 \in \mathcal{C}, \quad B_4 \in \mathcal{C} \cup \mathcal{T},$$

and the only critical transient satisfies

$$(B_2)_1 = 0, \quad (B_2)_0 \leq \tau(c).$$

Thus

$$B_2 \in \mathcal{E}(\tau(c)).$$

The same transport satisfies the local repair screen with exact common margin

$$\delta(c) = \frac{23}{128} - \frac{c}{4} > 0. \quad (7)$$

Proof. Because $b_1 = 0$,

$$q_0 = 0, \quad q_1 = \frac{c}{4}, \quad q_2 = \frac{c}{2}, \quad q_3 = \frac{c}{4} + \frac{1}{4}b_2.$$

We impose two structural zeros:

$$\pi(0, 1) = 0, \quad \pi(2, 3) = 0.$$

The first puts the $x = 0$ child in CORE; the second preserves $(B_2)_1 = 0$.

Put

$$r_2 = \left(q_1 + q_2 - \frac{13}{64} \right)_+,$$

$$r_3 = \left(q_1 + q_2 + q_3 - \frac{25}{64} \right)_+,$$

and set

$$r = \max\{r_2, r_3\}.$$

Assign $\pi(2, 2) = r$.

Column 1 is served by row 1 alone; this is possible because

$$q_1 = \frac{c}{4} \leq \frac{1}{8} < \frac{12}{64} = P_b(1).$$

After this, columns 1, 2 fit into rows 0, 1 plus the chosen mass r at $(2, 2)$ exactly when $r \geq r_2$. Likewise columns 1, 2, 3 fit into rows 0, 1, 3 plus the same mass r exactly when $r \geq r_3$. Thus the low-column flow exists by construction.

Since $b_2 \leq 1 - c$,

$$r_3 \leq \left(c + \frac{1-c}{4} - \frac{25}{64} \right)_+ = \left(\frac{48c-9}{64} \right)_+,$$

whereas

$$r_2 = \left(\frac{48c-13}{64} \right)_+.$$

Hence

$$r \leq \left(\frac{48c-9}{64} \right)_+.$$

For $c \leq 1/2$, this upper bound is at most $q_2 = c/2$, so the selected mass is available in column 2. Therefore

$$(B_2)_0 = \frac{r}{38/64} \leq \max \left\{ 0, \frac{48c-9}{38} \right\} = \tau(c),$$

and $(B_2)_1 = 0$ by the imposed zero at $(2, 3)$.

The $x = 0$ row avoids columns 0, 1, hence $B_0 \in \mathcal{C}$.

For row 4,

$$F_q(4) = c + \frac{3}{4}b_2 + \frac{1}{4}b_3 \leq \frac{3}{4} + \frac{c}{4} \leq \frac{7}{8}.$$

Thus at least $1/8$ of the mass lies in $u \geq 5$. If $q_5 \geq 1/64$, place row 4 at $u = 5$ and obtain t . Otherwise more than $7/64$ lies in $u \geq 6$, so row 4 can be placed there and yields a CORE child. After the low columns and row 4 are fixed, all remaining columns are at least 4 and can be filled by the residual capacities of rows 0, 1, 2, 3.

For the repair margin, from (3)

$$D_{21} - \frac{3}{128} = \frac{23}{128} - \frac{c}{4} = \delta(c).$$

Also

$$H_3(B) = c + \frac{1}{4}b_2 \leq \frac{1}{4} + \frac{3}{4}c \leq \frac{5}{8},$$

so $D_{43} \geq 23/64$ and its repair slack is at least $45/128$. Since $D_{\text{sh}} \leq 13/64$, the union slack is at least

$$\frac{21}{64} - \frac{c}{4},$$

which exceeds $\delta(c)$ by $19/128$. Thus the first donor inequality is the bottleneck and (7) is feasible. $\square$

7 Landing threshold

PROVED

Lemma 1 (Landing – PROVED). *If*

$$B \in \mathcal{E}\left(\frac{13}{48}\right),$$

then there exists a causal transport with

$$B_0 \in \mathcal{C}, \quad B_2 \in \mathcal{C}, \quad B_4 \in \mathcal{C} \cup \mathcal{T}.$$

Proof. Let $c = b_0$ and use $b_1 = 0$. Then

$$q_1 + q_2 = \frac{3}{4}c \leq \frac{13}{64}.$$

Serve column 1 with row 1 and columns 1, 2 with rows 0, 1, keeping $\pi(0, 1) = 0$. This forces $\pi(2, 2) = 0$, hence $(B_2)_0 = 0$.

Moreover

$$q_3 = \frac{c}{4} + \frac{1}{4}b_2 \leq \frac{1}{4},$$

so

$$(B_2)_1 \leq \frac{1/4}{38/64} = \frac{8}{19} < \frac{13}{16}.$$

Thus $B_2 \in \mathcal{C}$. The $x = 0$ and $x = 4$ arguments are the same as in the peeling theorem. $\square$

8 Exact hard-edge entry chain

PROVED

The boundary state is

$$G_0(A, C) = \frac{1}{2} + \frac{1}{2}A(t)C(t).$$

If

$$A(0) = C(0) = 0,$$

then the constant and linear coefficients of $A(t)C(t)$ are both zero. Hence

$$[G_0(A, C)]_{t^0} = \frac{1}{2}, \quad [G_0(A, C)]_{t^1} = 0,$$

so every such hard-edge product boundary state lies in $\mathcal{E}(1/2)$.

The scalar recurrence is now exact:

$$c_0 = \frac{1}{2},$$

$$c_1 = \tau(c_0) = \frac{15}{38},$$

$$c_2 = \tau(c_1) = \frac{189}{722}.$$

Furthermore

$$189 \cdot 48 = 9072 < 9386 = 13 \cdot 722,$$

therefore

$$\boxed{\frac{189}{722} < \frac{13}{48}}.$$

The next critical step is a landing step into CORE.

Corollary 1 (Three-step critical-channel entry – PROVED). *For every hard-edge product boundary state $G_0(A, C)$ with $A(0) = C(0) = 0$, there is a critical-channel graded transport path*

$$\boxed{\mathcal{E}\left(\frac{1}{2}\right) \xrightarrow{x=2} \mathcal{E}\left(\frac{15}{38}\right) \xrightarrow{x=2} \mathcal{E}\left(\frac{189}{722}\right) \xrightarrow{\text{landing}} \mathcal{C}.$$

At each transient stage, the $x = 0$ child enters CORE and the $x = 4$ child enters CORE or t .

The exact worst-case repair margins along this path are

$$\delta\left(\frac{1}{2}\right) = \frac{7}{128},$$

$$\delta\left(\frac{15}{38}\right) = \frac{197}{2432},$$

$$\delta\left(\frac{189}{722}\right) = \frac{5279}{46208},$$

followed by the CORE margin

$$\frac{23}{128}.$$

All are strictly positive.

9 Numerical regression layer

LP-CHECKED

This section is deliberately not part of the proof.

DREAM6-ZR-PHASE v0.2 used one coupled LP per semantic node: a root transport plus linked transports for the critical children $x \in \{0, 2, 4\}$. Its completed run had 119 unique laws, 100 LP-feasible states and 19 infeasible states. Ninety-nine of the 100 feasible states had positive common margin. The run reported no unknown or numerical-reject status. Its worst recorded equality residual was approximately 6.25×10^{-12} , with zero inequality violation and zero clipped negative transport mass.

The independent script `DREAM6_ZR_ENTRY_CERT_v01.py` was then used only as a crash-test of the exact constrained transport constructions above. The final regression consisted of 310 extreme one-atom-tail checks and 10,000 random finite-support laws, through tail index 32. With explicit HiGHS feasibility tolerances, the final run returned PASS with maximum equality residual approximately 4.44×10^{-16} and zero inequality violation.

An earlier run under looser/default numerical settings produced one apparent failure with equality residual about 6.3×10^{-8} . Tightening the LP feasibility tolerance removed it. This incident is retained in the ledger because it illustrates the correct epistemic status:

LP PASS is regression evidence, not a mathematical certificate.

In particular, no floating-point PASS is used to establish any proposition, lemma, theorem, or corollary labeled PROVED above.

10 What remains open

OPEN

The local critical-channel program is not the full causal theorem. The following gates remain.

1. Branches $x = 1, 3$

A full five-child causal witness has branches 0, 1, 2, 3, 4. The exact entry grammar above controls only the critical channels 0, 2, 4. Branches 1, 3 must be incorporated into a well-founded recursion that tracks the factor depths consumed by those calls. No such five-branch induction is proved here.

2. Terminal fuel

The law t is a base/terminal object, not an indefinitely viable invariant state. A transition into t is valid only when the remaining depth/fuel label makes that branch terminal. The local transport proof identifies where t may arise; it does not by itself certify the required remaining-depth semantics.

3. Ordered-history lift

The coarse transport quotient forgets ordering/history information. A full causal construction must lift the graded coarse transport to ordered histories while preserving positivity and the required compatibility across histories. The six-dimensional local phase tells us what the boundary can see; it does not prove that the phase composes upstairs.

4. Global consequences

Until the preceding three gates close, the following statements remain open:

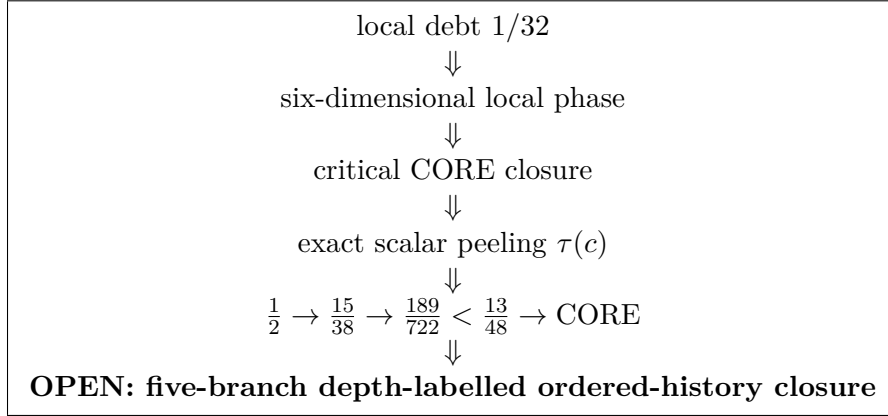
full causal addition lemma,

full boundary closure,

$O(\log n)$ upper bound.

11 Addendum-ready conclusion

The correct concise status is:



The checkpoint therefore establishes a finite exact critical-channel entry mechanism. It does not close the causal addition theorem. The next proof obligation is not another local LP: it is the well-founded five-branch, depth-labelled, ordered-history lift.